

EXPERIMENT - 16

1. Problem Statement

Write a menu-driven C program using `switch-case` to perform digit-based operations on a positive integer. Define one separate user-defined function for each menu option, with void return type and with a simple function call (without call by value/reference). Each function should read the required input, process it, and display the result.

- Check whether its digits are in increasing order from left to right.
- Check whether its digits are in decreasing order from left to right.
- Find the largest digit and the smallest digit.
- Find the second largest digit and second smallest digit in the number.

2. Algorithm

Step 1 : START

Step 2 : Display the menu

1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit

Step 3 : Read choice

Step 4 : switch (choice)

case 1 : call check_increasing()	break
case 2 : call check_decreasing()	break
case 3 : call largest_smallest()	break
case 4 : call second_largest_smallest()	break
default: Print "Invalid choice"	break

Step 5 : STOP

FUNCTION check_increasing()

```
Read num
If num <= 0 then Print error and return
Set flag = 1, temp = num
prev = temp % 10
temp = temp / 10
While temp > 0 repeat
    cur = temp % 10
    If cur >= prev then flag = 0
    prev = cur
    temp = temp / 10
End While
If flag = 1 then Print "in increasing order"
Else Print "not in increasing order"
```

FUNCTION check_decreasing()

```
Same as above, only the condition changes to
If cur <= prev then flag = 0
and the messages change to decreasing order
```

FUNCTION largest_smallest()

```
Read num
If num <= 0 then Print error and return
Set largest = 0, smallest = 9, temp = num
While temp > 0 repeat
    digit = temp % 10
    If digit > largest then largest = digit
    If digit < smallest then smallest = digit
    temp = temp / 10
End While
Print largest and smallest
```

```
FUNCTION second_largest_smallest()
    Read num
    If num <= 0 then Print error and return
    Set largest = -1, second_largest = -1
    Set smallest = 10, second_smallest = 10
    temp = num
    While temp > 0 repeat
        digit = temp % 10

        If digit > largest then
            second_largest = largest
            largest = digit
        Else If digit < largest and digit > second_largest then
            second_largest = digit
        End If

        If digit < smallest then
            second_smallest = smallest
            smallest = digit
        Else If digit > smallest and digit < second_smallest then
            second_smallest = digit
        End If

        temp = temp / 10
    End While
    If second_largest = -1 then Print "does not exist"
    Else Print second_largest
    If second_smallest = 10 then Print "does not exist"
    Else Print second_smallest
```

3. Test Case Table

TC #	Choice	Number	Expected Output	Actual Output	Result
TC1	1	1234	Digits are in increasing order	Digits are in increasing order	PASS
TC2	1	1325	Digits are not in increasing order	Digits are not in increasing order	PASS
TC3	1	1223	Digits are not in increasing order	Digits are not in increasing order	PASS
TC4	1	7	Digits are in increasing order	Digits are in increasing order	PASS
TC5	2	9631	Digits are in decreasing order	Digits are in decreasing order	PASS
TC6	2	9361	Digits are not in decreasing order	Digits are not in decreasing order	PASS
TC7	2	5541	Digits are not in decreasing order	Digits are not in decreasing order	PASS
TC8	3	5382	Largest = 8 Smallest = 2	Largest = 8 Smallest = 2	PASS
TC9	3	1020	Largest = 2 Smallest = 0	Largest = 2 Smallest = 0	PASS
TC10	3	777	Largest = 7 Smallest = 7	Largest = 7 Smallest = 7	PASS
TC11	4	5382	Second largest = 5 Second smallest = 3	Second largest = 5 Second smallest = 3	PASS
TC12	4	4455	Second largest = 4 Second smallest = 5	Second largest = 4 Second smallest = 5	PASS
TC13	4	10	Second largest = 0 Second smallest = 1	Second largest = 0 Second smallest = 1	PASS
TC14	4	777	Both do not exist	Second largest : does not exist Second smallest : does not exist	PASS
TC15	1	-5	Error message	Error: -5 is not a positive integer.	PASS
TC16	3	0	Error message	Error: 0 is not a positive integer.	PASS
TC17	7	-	Invalid choice message	Invalid choice! Enter 1 to 4.	PASS

4. Source Code

File name: `digit_menu.c` (209 lines)

```
/* HDOC Day-16
Menu-driven program using switch-case to perform digit based
operations on a positive integer.
One separate user-defined function is written for every menu
option. All the functions have void return type and are called
by a simple function call, without any parameter. */

#include <stdio.h>

void check_increasing(void);
void check_decreasing(void);
void largest_smallest(void);
void second_largest_smallest(void);

int main(void)
{
    int choice;

    printf("===== MENU =====\n");
    printf("1. Check whether digits are in increasing order\n");
    printf("2. Check whether digits are in decreasing order\n");
    printf("3. Find the largest digit and the smallest digit\n");
    printf("4. Find the second largest and second smallest digit\n");
    printf("===== \n");
    printf("Enter your choice (1 to 4): ");
    scanf("%d", &choice);

    switch (choice) {

        case 1:
            check_increasing();
            break;

        case 2:
            check_decreasing();
            break;

        case 3:
            largest_smallest();
            break;

        case 4:
            second_largest_smallest();
            break;

        default:
            printf("\nInvalid choice! Enter 1 to 4.\n");
            break;
    }

    return 0;
}
```

```

/* Option 1 : digits in increasing order from left to right */
void check_increasing(void)
{
    long num, temp;
    int prev, cur, flag;

    printf("\nEnter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("\nError: %ld is not a positive integer.\n", num);
        return;
    }

    flag = 1;
    temp = num;
    prev = temp % 10;
    temp = temp / 10;

    while (temp > 0) {
        cur = temp % 10;
        if (cur >= prev)
            flag = 0;
        prev = cur;
        temp = temp / 10;
    }

    printf("\nNumber : %ld\n", num);
    if (flag == 1)
        printf("Output : Digits are in increasing order\n");
    else
        printf("Output : Digits are not in increasing order\n");
}

/* Option 2 : digits in decreasing order from left to right */
void check_decreasing(void)
{
    long num, temp;
    int prev, cur, flag;

    printf("\nEnter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("\nError: %ld is not a positive integer.\n", num);
        return;
    }

    flag = 1;
    temp = num;
    prev = temp % 10;
    temp = temp / 10;

    while (temp > 0) {
        cur = temp % 10;
        if (cur <= prev)
            flag = 0;
        prev = cur;
        temp = temp / 10;
    }
}

```

```

    printf("\nNumber : %ld\n", num);
    if (flag == 1)
        printf("Output : Digits are in decreasing order\n");
    else
        printf("Output : Digits are not in decreasing order\n");
}

/* Option 3 : largest digit and smallest digit */
void largest_smallest(void)
{
    long num, temp;
    int digit, largest, smallest;

    printf("\nEnter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("\nError: %ld is not a positive integer.\n", num);
        return;
    }

    largest = 0;
    smallest = 9;
    temp = num;

    while (temp > 0) {
        digit = temp % 10;
        if (digit > largest)
            largest = digit;
        if (digit < smallest)
            smallest = digit;
        temp = temp / 10;
    }

    printf("\nNumber          : %ld\n", num);
    printf("Largest digit   : %d\n", largest);
    printf("Smallest digit  : %d\n", smallest);
}

/* Option 4 : second largest digit and second smallest digit */
void second_largest_smallest(void)
{
    long num, temp;
    int digit;
    int largest, second_largest;
    int smallest, second_smallest;

    printf("\nEnter a positive integer: ");
    scanf("%ld", &num);

    if (num <= 0) {
        printf("\nError: %ld is not a positive integer.\n", num);
        return;
    }

    largest = -1;
    second_largest = -1;
    smallest = 10;
    second_smallest = 10;
    temp = num;

```

```

while (temp > 0) {
    digit = temp % 10;

    if (digit > largest) {
        second_largest = largest;
        largest = digit;
    }
    else if (digit < largest && digit > second_largest) {
        second_largest = digit;
    }

    if (digit < smallest) {
        second_smallest = smallest;
        smallest = digit;
    }
    else if (digit > smallest && digit < second_smallest) {
        second_smallest = digit;
    }

    temp = temp / 10;
}

printf("\nNumber          : %ld\n", num);

if (second_largest == -1)
    printf("Second largest digit : does not exist\n");
else
    printf("Second largest digit : %d\n", second_largest);

if (second_smallest == 10)
    printf("Second smallest digit : does not exist\n");
else
    printf("Second smallest digit : %d\n", second_smallest);
}

```

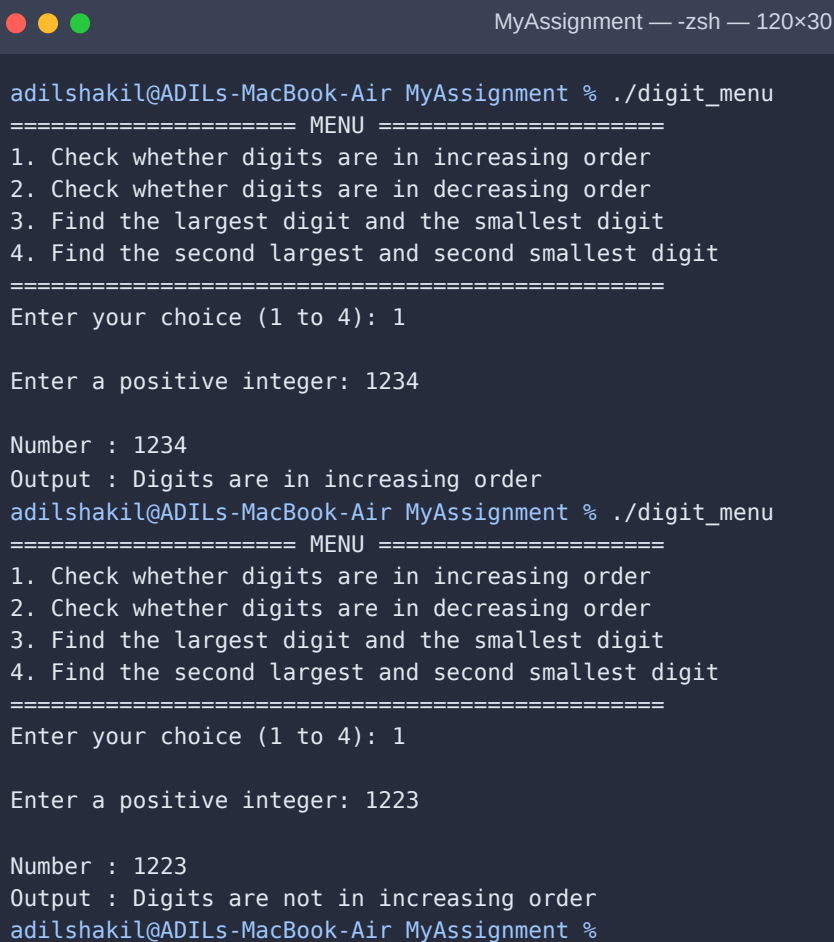

5. Compilation and Output

I wrote the program in the `MyAssignment` folder using the `nano` editor, then compiled and ran it:

```
$ cd ~/Desktop/MyAssignment
$ nano digit_menu.c
$ gcc digit_menu.c -o digit_menu
$ ./digit_menu
```

The program compiled in the first attempt itself, without any error or warning.

Option 1 : increasing order



```
MyAssignment — zsh — 120x30

adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
===== MENU =====
1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit
=====
Enter your choice (1 to 4): 1

Enter a positive integer: 1234

Number : 1234
Output : Digits are in increasing order
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
===== MENU =====
1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit
=====
Enter your choice (1 to 4): 1

Enter a positive integer: 1223

Number : 1223
Output : Digits are not in increasing order
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Option 2 : decreasing order

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
===== MENU =====
1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit
=====
Enter your choice (1 to 4): 2

Enter a positive integer: 9631

Number : 9631
Output : Digits are in decreasing order
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
===== MENU =====
1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit
=====
Enter your choice (1 to 4): 2

Enter a positive integer: 9361

Number : 9361
Output : Digits are not in decreasing order
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Option 3 : largest and smallest digit

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
===== MENU =====
1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit
=====
Enter your choice (1 to 4): 3

Enter a positive integer: 5382

Number      : 5382
Largest digit : 8
Smallest digit : 2
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
===== MENU =====
1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit
=====
Enter your choice (1 to 4): 3

Enter a positive integer: 1020

Number      : 1020
Largest digit : 2
Smallest digit : 0
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Option 4 : second largest and second smallest digit

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
===== MENU =====
1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit
=====
Enter your choice (1 to 4): 4

Enter a positive integer: 5382

Number          : 5382
Second largest digit : 5
Second smallest digit : 3
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
===== MENU =====
1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit
=====
Enter your choice (1 to 4): 4

Enter a positive integer: 4455

Number          : 4455
Second largest digit : 4
Second smallest digit : 5
adilshakil@ADILs-MacBook-Air MyAssignment %
```

All digits same, and the invalid cases

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
```

```
===== MENU =====
```

1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit

```
=====
```

```
Enter your choice (1 to 4): 4
```

```
Enter a positive integer: 777
```

```
Number : 777
```

```
Second largest digit : does not exist
```

```
Second smallest digit : does not exist
```

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
```

```
===== MENU =====
```

1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit

```
=====
```

```
Enter your choice (1 to 4): 1
```

```
Enter a positive integer: -5
```

```
Error: -5 is not a positive integer.
```

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./digit_menu
```

```
===== MENU =====
```

1. Check whether digits are in increasing order
2. Check whether digits are in decreasing order
3. Find the largest digit and the smallest digit
4. Find the second largest and second smallest digit

```
=====
```

```
Enter your choice (1 to 4): 7
```

```
Invalid choice! Enter 1 to 4.
```

```
adilshakil@ADILs-MacBook-Air MyAssignment %
```